

Prologue to the 20th Indian Antarctic Expedition

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Leader

Introduction

I was asked to lead the 20th Indian Antarctic Expedition just a month before the Expedition sailed. I was slightly apprehensive about the team as I had not interacted with any of the members. I accepted the challenge and the outcome was rewarding. I met my team at NCAOR, Goa four days before it sailed. It was a good team and I was confident of having a very successful Expedition to Antarctica. Today I feel satisfied for having led a very good team that thrived at every step to achieve in true spirit of companionship of an Expedition.

The Expedition

The XX Indian Scientific Expedition to Antarctica was launched from Dabolim Airport, Goa on 29th Dec 2000. The Indian Team of 52 members was reduced to 51 members as Shri S.K. Joshi from SASE, Manali had to be left behind on medical grounds. The entire team moved to Mumbai and then to Cape Town, South Africa. Bulk baggage booking at the airport was done so that the team could reach Cape Town without incurring any additional cost for carrying official baggage and equipment like Inmarsat Dome (Radoom), books, medicines, GPR equipment etc. The team on reaching Cape Town at 1200 hrs (LST) had to be moved to the port to board the ship M V Magdalena Oldendorff as the Expedition was Scheduled to depart on 30th Dec 2000. In the evening the team attended a Farewell function organized by Indian Conciliate at Cape Town and NCAOR, Goa. At an impressive function the Indian Antarctic Team was given send off by the High Commissioner of India to South Africa. Also present at the function were Consul General of Indian consulate at Cape Town Mr. Puri and NCAOR Center Director Dr. P. C. Pandey.

The Indian Expedition sailed from Cape Town aboard ship M/V Magdalena Oldendorff at 2330 hrs on 30-12-2000. Two members from

Technical University Dresden and Four-member helicopter crew of Helicopters (NZ) Ltd from New Zealand joined the expedition at Cape Town. The expedition carried 11 containers of general stores, spares, food and medicines and two refrigerated containers of fresh and frozen food. The fuel carried by the expedition included 270 KL of bulk fuel, 30 KL of fuel in epoxy coated barrel and lubes. The Polar vehicle Polar Bear was also taken back to Antarctica after repairs.

The ship crossed 40° S. lat. on 1st Jan 2001. The sailing to Antarctica was extremely good (Fig. 1) and the expedition encountered rough sea only for about 20 hrs. However, very dense pack ice was encountered in the morning of 8th January 2001 at 67° S. lat. and 9° 52'E. long. The progress of ship was hindered due to the pack ice and could reach up to 68° 47' S. lat. and 10° 15' E. long, in the late morning of 9th January 2001, which was about 120 NM from Maitri. On 9th January 2001, the ship was instructed to move in the vicinity of India Bay i.e. at 12° E Long, from its position at 9°45' E. Long. The weather was not appropriate for Heli-operations on 9th, 10th and 11th Jan 2001. The ship reached 12° E. Long, on 12th morning and started moving south as it got some opening and was placed about 70 NM from Maitri. It could not go any farther as the entire India Bay and adjoining Lazarev Bay and Russian Bay were covered with fast ice. The first landing at Maitri station was made in the forenoon of 12th Jan 2001. Shifting of cargo and Scientific personnel started on 12th January 2001 and by the 14th Jan 2001, all the cargo related to scientific work and the satellite communication equipment carried by 20th IAE was shifted to Maitri. This cargo also included the 3 ton equipment brought by TU Dresden German Group. The entire scientific group along with 5 logistic personnel was shifted to Maitri. The field camps for Survey of India and the Wadia Institute of Himalayan Geology were established by the 16th January 2001. The tentages for IIG camp at the nunatak Sigurd Svodene were placed at the site on 18th January 2001 and this camp was made functional on 21st January. The ship was completely vacated except for the NHO team and the Doctor by 18th January 2001. The Helicopters were, for the first time, shifted to Maitri and parked there for 16 days, as the entire activity of the expedition was centered at Maitri. This resulted in planning the field activities without wasting the flying time between Maitri and ship. The logistic team that was moved to Maitri started in right earnest to interact with the 17th WOT. Shri Arun Chaturvedi, Station Commander Maitri was more than willing to extend all the help he could to make us feel at home in Maitri. Logistic team had a very fruitful on job training on all systems at Maitri. On advice from Shri Aran Chaturvedi, we decided to take over Maitri station only when my team was fully

confident of running the station flawlessly. The station Maitri was finally handed over to **the** 18th WOT on 18th February 2001. We soon realized that manning the Station Maitri was a matter of honor to all of us and we resolved to do our best for up-keeping India's symbol of pride in Antarctica. We quickly learnt the basics of manning Maitri and put in all our effort in the year to come to ensure that Maitri along with its surroundings was maintained without disturbing the fragile Antarctic Environment. Meanwhile a tricky situation had developed in the India bay region as the fast ice surrounding it had not broken as a result of which the ship was unable to berth and discharge the cargo on to the shelf. It was only on 1st March the ship could berth along the shelf but had to move out immediately due to bad weather. The ship came back to the shelf on 3rd March. We quickly and safely completed the offloading operation. The Ship Magdalena Oldendorff finally departed from Antarctic shore leaving **the** 18th WOT to keep the Indian flag flying high at the Indian Antarctic Base 'Maitri'. That's when our Wintering began.

The Wintering

Wintering is a one long experience very few men/women try and endure. We were amongst fortunate few **who** could experience the Antarctic wilderness. Our routine was quickly set with Convoys, station maintenance work and scientific commitment. The months of March-April were perfect to do a lot of initial outdoor work around the station. This included arranging the stores and supply that had come with us and also carrying out maintenance of life support system to sustain us through the two months of darkness. Slowly the blizzards became more frequent. Blizzards required additional effort as they choked the generators and part of our living area like the toilets with drift snow. It was always a huge task to clear the snow after and during the blizzard from the passage and generator rooms. In May we had a rather nasty convoy to the shelf. It was stuck in a blizzard and could return only after 14 days. We at the station had some anxious moments during this period as the communication link with the convoy was not perfect. The convoy had to be called back from DG station area. The return journey was particularly eventful as three snow vehicles developed snags and were abandoned en-route to Maitri only to be salvaged in September 2001. The Sun finally set for two months on May 22nd. The dark Antarctic nights with spectacular auroral displays were an everyday feature now. During this period we confidently embarked upon the task of changing the station heating pipeline. It was a total teamwork that helped station Engineer and his team to change it within a span of 24

hours without any flaws. The station heating was perfect now for the cold Antarctic winter. The blizzards during this period made our task more demanding. The team spirit was at its peak and I decided to cash on it. We took up the task of repainting the station lobby of Maitri station. This area was a veritable museum of plaques and boards. We converted it into a gleaming lobby. We also repainted the lounge and Kitchen.

The Sun was up again in July. Very characteristically I could notice a slight dip in the spirit of my team mates. Antarctic isolation was catching up. But with daylight our commitments increased. Unusual snow accumulation had occurred all around. The vehicles had to be removed from their snow graves. The container area had to be cleared and the vehicles serviced for post winter task. We also took up the task of painting the Station exterior with the very little paint that we had. Cleaning of Maitri and its surrounding became an important task. The vehicles left on Ice Shelf were retrieved. We had a busy schedule and that perhaps kept the isolation blues at bay. We had a big Diwali celebration. November is always a perfect month in Antarctica. I was out for field work quite often. In early December the Flight from Cape Town landed at the Russian air strip south of Maitri. We got our supply of fresh vegetables and that cheered all. Christmas time was a celebration time again. We had our Russian friends with us on this occasion. The year 2002 was approaching and that's when we thought we should be going home. An eager wait for the new team began. On 18th January 2002 as we welcomed Dr P C Pandey, former Director, NCAOR and R P Lai, leader 22ⁿ Expedition our long wintering in the Icy wilderness had come to an end. I personally think that we did our best to make the Indian Antarctic base a better place to live and work in. We had also taken all care to preserve the fragile ecosystem of Antarctica. The logistic task we completed has been elaborated in the report on logistics by Maj. Atul Apte.

An attempt is made here to give in brief the outcome of scientific task assigned to the 20th Expedition.

Scientific Task—Summer

A. Atmospheric Sciences and Meteorology

Geomagnetic Studies by Dr. S. Gurubaran, M. Sridharan, K. Jeeva and Sumendra Singh of Indian Institute of Geomagnetism (IIG), Mumbai

Objectives

1. Monitoring of the Magnetosphere-Ionosphere interactions using Fluxgate Magnetometer and Riometer.

2. Measurement of Total Geomagnetic Field (F) using Proton Precession Magnetometer.
3. Measurements of atmospheric electrical parameters in connection with the Global Electric Circuit using Maxwell current sensor and **Field Mill**.

Outcome

1. The 3-station triangulation experiment involving the fluxgate magnetometers was carried out during the Antarctic summer month of January 2001. At Maitri (70° 45'S, 11°44'E and Dakshin Gangotri (70°05'S, 12°E), the experiment was initiated on 18 January and was continued till 30 January. At the third station (71°20'S, 7°38'E) near Sigurd Mountain, IIG scientists operated the magnetometer for six days during 23-28 January 2001. Both DV (diurnal variation) and magnetic pulsations were recorded and the data are available in both digital and analog forms. At DG the sensor alignment was found disturbed on 24 January and the same was corrected. The DV chart recorder at DG started malfunctioning in the midst of the campaign. There were blizzard-like conditions at DG between 26 and 28 January and this has contributed to the break in the analog records. At the mountains continuous digital data were recorded between 24 and 27 January. Due to the malfunctioning of the generator, the experiment had to be terminated on 28 January. There were quite a few events involving magnetic disturbances (both substorm-like signatures and bays) during the period (24-26 January) for which data (analog and/or digital) from all the three stations are available.
2. A new electrometer was incorporated in the Maxwell current experimental setup near Kamet hut and this enabled representation of the sensor output in terms of atmospheric current density. A new antenna wire was erected towards the end of the summer campaign and its performance was found satisfactory.
3. The Electric Field Mill was commissioned successfully near the Kamet hut and the experiment was conducted for more than four weeks. Reliable data on the atmospheric electric field were obtained during the period 16 January - 24 February.

Summary

1. The 3-station triangulation experiment was carried out during the Antarctic summer of 2000-2001. Simultaneous magnetic records (digital and/or analog) for more than three days are available during

this campaign period (23-28 January 2001). Their analysis is expected to yield useful information on the dynamics of the auroral current systems.

2. Magnetic records from Maitri representing intense substorm events (for example, the event on 13-14 February 2001) will be analyzed in conjunction with several other measurements available elsewhere in order to throw more light on the intensification of auroral current systems at subauroral latitudes and the occurrence of Harang discontinuity.
3. The decline in the geomagnetic field as detected in earlier measurements from Maitri will continue to receive attention in one of the ongoing studies that will involve data collected during the XX Indian Antarctic Expedition.
4. Surface measurements of atmospheric electrical parameters carried out near the Kamet hut at Maitri provide excellent scope for carrying out studies related to the global electric circuit. A maximum at 2000 UT observed on many days conforms to the traditional view that the thunderstorms are the main drivers of the global circuit. Effects of local meteorological conditions on the measured electrical parameters and the magnetospheric contributions to the ionospheric potential and the global circuit will be investigated in detail in the forthcoming study.

Climatological and Meteorological studies by S. Stella, Suresh Ram and M. K. De of India Meteorological Department, N. Delhi.

Objectives

1. Collection of Meteorological data of Indian Antarctic Station Maitri.
2. Measurement of total ozone, damaging UV (D-UV), NO_2 , using Brewer Spectrophotometer.
3. Vertical profile of ozone concentration using ozone sonde ascents.

Outcome

1. Continuous observation of prevailing weather over and around Maitri.
2. Continuous recording of air temperature, wind direction and speed, atmospheric pressure, surface ozone and global solar radiation have been made.

3. Recording of 3 hourly synoptic observations (00,03,06,09,12,15,18 & 21 UTC).
4. Brewer had been operated for 19 days in Jan '01 and 13 days in Feb '01 under favorable sky condition for operation. To measure Total ozone, D-UV, N02
5. Four ozone sonde ascents were taken during summer period. All the ascents reached more than 30 kms height.
6. Measurement of atmospheric turbidity was done using Sun Photometer with filters.

Besides this IMD also provided Meteorological services to other participants and issued daily weather forecast for helicopter operations in Antarctica.

VLF Propagation studies by R.P. Patel of Physics Department, Barkatullah University, Bhopal.

Objective

1. Continuation of ongoing experiment of monitoring Very Low Frequency (VLF) radio wave propagation with a view to studying the Ionosphere and Magnetosphere interactions emanating from outer space.
2. To acquire the data for long distance communication study.
3. The study of directionality of whistlers (VLF).

Outcome

1. Repairing of old antenna was attempted.
2. Maintenance service of amplifier and recorder were done.
3. Data were acquired for four days using old antenna.
4. A suitable point for new pole was selected to acquire 40m length of antenna.
5. A 10m pole was erected near the Annapurna hut.
6. Using old and new poles the T' type antenna was successfully installed
7. Data were recorded continuously using digital audio recorder PCM-R300.
8. Manual detection of the whistler and related events was done from the recorded data.

B. Earth Sciences

Geological mapping and glaciological studies by Jayapaul D, Arun Chaturvedi, M. J. D'Souza and Amit Dharwadkar of Geological Survey of India, Antarctica Division, Faridabad.

Objective

1. Geological mapping on a scale of 1:50000 in eastern part of Muhlig-Hoffmannfjella between Long. 6°00' and 7°00'E and Lat. 71° 40' and 72°00'S.
2. Glaciomorphological mapping in the eastern part of Schirmacher Oasis on 1:25000 scale.
3. Monitoring accumulation of snow over Ice shelf through stake network.
4. Monitoring fluctuations of the polar ice margin.
5. Iceberg monitoring (on board ship during journey).

Outcome

1. **Geological Studies:** Geological mapping was carried out in the eastern part of Muhlig-Hoffmannfjella between Long. 6° 20' & 7° 7'20"E and Lat. 71°40'00" & 72° 05'S. In addition to the above, nunataks to the NE of Muhlig-Hoffmannfjella viz Drygalski and Sigurd-Svodene were also mapped on 1:50,000 scale and an area of 1200 sq.km was covered and 107 samples were collected for further studies.
2. **Glaciological Studies:** Glaciomorphological studies were carried out in the eastern part of Schirmacher Oasis
3. Monitoring of Dakshin Gangotri Glacier Snout to record the temporal and secular fluctuation in the polar ice front.
4. Icebergs were monitored during the voyage to Antarctica.
5. Monitoring of annual accumulation/ablation over the shelf ice through measurements of the stakes fixed on the Indian Bay shelf near D.G. camp was carried out.

Tele Seismic studies by Prem Kishore and S. Padhy of National Geophysical Research Institute, Hyderabad

Objective

The Seismic station established at Maitri will monitor the seismic activity within Antarctica. The station also aims at monitoring the tele seismic activity as well as the activity in mid oceanic ridge in the Indian Ocean. Data exchange with other stations besides participating in the global earthquake epicentral determination is envisaged.

Outcome

The seismic station is equipped with high-resolution digital seismograph station as well as analog seismic recorder. During austral summer the station was upgraded with the installation of additional Broad Band Seismic System with Gurlap CMG-3ESP sensor and Digital acquisition system RT 121 from Reftek. The seismic vault was opened for the installation of the Broad Band sensor and found intact without any droplet of ice inside. The recent Bhuj earthquake was recorded at Maitri seismic station and after analyzing the data the records were sent to NGRI through FAX. Using SEISAN software preliminary analysis of both the digital and analog records was carried out. Archiving of the DAT tapes as well as CDs for further analysis of the events was also done.

Planetary Geodetic Studies by Dr. M. Scheinert of TU Dresden, Germany in collaboration with NGRI, Hyderabad

Objective

1. Application of GPS to study geodynamics, especially vertical deformations of the earth's crust. The continental ice acts as a surface load on the solid earth and causes a viscoelastic response e.g. uplift when ice load decreases. The goal is to determine height difference for an epoch difference of about five years. Expectedly these height difference yield to a derivation of the vertical deformation pattern as a basis for further interpretation of historic and recent mass imbalances and a comparison with viscoelastic models for this region. For this three groups set in Gruber-Petermann, Humboldt-Insel and Orvin Mountains are to be reobserved for minimum of 5 days at each point with continuous GPS receiver base at Schirmacher Oasis.
2. Application of GPS for Glaciological Research/ground truthing required for remote sensing satellite. The objective is to provide base for ice mass balance of Antarctica especially for coastal region. The data thus collected will be validated by satellite and

can also be used to validate future satellite data independently. The observations for this will be made along traverse route to Untersee lake in Gruber. In this area the reference of three existing corner reflectors is to be checked and determined again.

Outcome

The three groups of GPS points were re-observed as projected for period of 5 to 6 days. The GPS observation for reference was taken for the entire period.

The traverse to Untersee was done using Skidoo with helicopter back up to set camp at Untersee. All the anticipated observations were made.

Summary

The data generated is expected to yield accurate information on surface geometry and surface height ranges as well as surface motion. The computation of this data will be used as input for computation concerning the height changes of bedrock elevation. The remote sensing data analysis will be performed in collaboration with Indian scientists and ground truth data will be used for data validation. The GPS station established by NGRI at Maitri enabled its participation and contribution of GPS data as a fundamental marker to International scientific committee of Antarctic Research (SCAR) GPS campaign for 2001. Data collected for the campaign has been converted to the standard RINEX format and stored on to a CD by NGRI for use by TU Dresden team.

Crack Propagation studies by Col. P. Mathur and Mahender Kumar of Snow and Avalanche Study Establishment, Manali

Objective

1. Energy budget on blue ice (continental ice) and Shelf ice an ongoing study
2. Crevasse study
3. Crack propagation study

Outcome

1. Collection of Albedo values and other parameters near Veteheia hill and Daksin Gangotri station done. Also at these locations Automatic Weather Stations with sensors recording wind speed and wind direction, relative humidity and ambient temperature and

atmospheric pressure were made operational along with Albedometer and snow depth sensors.

2. The snow grain identification studies indicated presence of predominantly sugary grains at Veteiah, Shivling and Russian Aerodrome site.
3. Numbers of crevasses have been identified between Veteheia hill and Shivling glacier slopes. These crevasses are developed in both transverse and longitudinal directions. Due to unusual weather condition and heavy drift the crevasses remained buried under drifting snow and could not be monitored. However GPS positions for these crevasses have been taken for future reference.
4. Crack propagation study of the crack in India bay has been initiated. GPS reading for the crack taken for long-term monitoring. NCAOR is requested to provide old photographs/satellite imageries of India bay

Ground Probe Radar Survey by Dr J. T. Gergen, Dr. D.P. Dhobal and Dr. R. J. Thayeen of Wadia institute of Himalayan geology, Dehradun.

Objective

1. Carry out GPR Survey on Polar Ice Sheet south of Schirmacher Oasis and on Ice Shelf in Dakshin Gangotri area

Outcome

1. An 8.3 Km long GPR profile between nunatak Veteheia and Tallaksenvadren was completed after putting in continuous efforts in field for more then 15 days. The elevation data for the profile was obtained with the help of Scientists from NHO, Dehradun.
2. Grid survey to select suitable ice core site was carried out in the area north of nunatak Tallaksenvadren.
3. Carried out GPR profiling to locate DG station and hanger buried under the snow over the years in Dakshin Gangotri station area.

Summary

1. The Ground Penetrating Radar Survey was carried out by 2.4 ns pulse console having 12.5 MHz centran frequency Antennae which were aligned longitudinal to the profile line. The sounding was performed with a 2m step size near the Veteheia nunatak (Lat

70°47.75' S and Long 11° 41.73'E). The ice thickness at Veteheia was 5-7 meters which gradually increased to 100m at 400 m along the survey profile line due south. The bedrock profile till 900m is seen to be sharply dipping which subsequently flattens with small undulations till 3100m. From this point bedrock profile again attains a gentle slope and at 4760 m it gets out of pulse EKKOIV 2.4ns console.

2. The grid survey for the Selection of ice core drill site is performed by 12.5MHz antenna. A Common Mid Point (CMP) was also conducted over this grid using 12.5 MHz antenna. Some buried object was detected on the northern corner of the grid, underneath a Russian wooden pole. The bedrock at this location is out of range of the GPR time window. The profile sounded between Tallaksenvadren and the grid site show that the ice thickness reaches 450m at 700m away from the Tallaksenvadren nunatak.

Three profiles were sounded at DG. The one taken to study the reflection pattern from sea water- ice shelf interface indicated a thickness of 175-180m of ice shelf. The reflections from the seawater are very prominent and continuous. In the second profile very strong reflections from the buried DG station were picked up and the third one located the hangar, which is now buried under 12-15m of snow.

Topographic mapping by Survey of India, Dehradun

Objective

1. Detail topographic survey on 1:1000 scale and at 1 meter contour interval for an area of 500 x 400 sq. meters, in Schirmacher Oasis.

Outcome

The given assignment was carried out in two parts i.e. Control Extension and Plane Table Survey.

Control Extension

1. A traverse was run based on the cassini coordinates of Maitri and Trishul station.
2. Total 10 traverse stations were made during the progress of the work.
3. 20 offsets were also made in and around area for densifying the control points.

4. The computations were made to find out the coordinates and heights of all the traverse stations and offsets.

Plane Tabling Survey

1. Normal Survey of India procedure was adopted for Plane Table Survey at 1:1000 scale and at 1 meter Contour Interval.
2. The calculated coordinates of traverse stations and offsets in and around the area to be surveyed were plotted on the Plane Table section.
3. The Contouring was done by Clinopole chasing.
4. Total 40 fixings were made in the area for detailed field survey. A Total of 500m x 400 m area was surveyed.

Hydrographic Survey by Lt. Cdr T P Mahato and Avijit Nag of Naval Hydrographic Survey team

Objectives

1. Collection of bathymetric data for eventual publication of a navigational chart, at a scale of 1:200,000 in an area bounded by 69 deg 24 min S to 70 deg 47 min S and 11 deg to 14 deg E. The aim was to collect bathymetric data as much possible since the area is large and cannot be completed in a single season.
2. Collection of Physical Oceanographic data in the area of survey.
3. Demarcations of the ice shelf falling in the area (11 deg to 14 deg E).
4. Collection of meteorological data.

Outcome

1. Bathymetry: Unlike other years, the area was under ice cover for most of the duration of the expedition. The fast ice in India Bay area cleared only after 21 February. The ship was then required to berth along the shelf for cargo offloading. The bathymetric survey thus could not be completed.
2. Physical Oceanographic data: At three locations, current directions and speed, water temperature, salinity and velocity of sound were observed using Smart acoustic Current meter for duration of 17 hrs, 12 hrs and 3hrs.

3. Ice shelf delineation: Ice shelf delineation was undertaken on 27 Feb and 4th March 2001. Ice shelf of about 100 kms (from 11 deg to 12 deg 56 min E) was delineated using the Helicopter. The positioning was done by flying over the ice shelf using Leica GPS 200. The data obtained is satisfactory and meets accuracy standards.
4. Meteorological data: Meteorological data comprising of atmospheric temperature, atmospheric pressure, sea surface temperature, cloud, sea state, wind direction and force has been collected during the stay at the ship. The data would be used in database and correlating with other collected data.
5. The teams experience in handling GPS positioning was used by WIHG, BSIP, SASE and Goa University.

C. Global Change

Palyno-stratigraphic studies by Dr S K Bera of Birbal Sahni Institute of Palaeobotany (BSIP), Lucknow.

Objective

1. The study of aerospora and surface sediments distributed in and around Schirmacher oasis in order to establish a pollen-spore deposition model in Antarctica.
2. The study of glacial lake sediments in order to calibrate the events and episodes leading to palaeoclimatic interpretation in Antarctica.

Outcome

1. Collection of air samples using Burkard Volumetric Sampler (UK) during voyage starting from 40 deg. South latitude to Indian Research Station Maitri, Schirmacher Oasis, East Antarctica.
2. Thirty five moss cushions, 15 dry soil samples, 5 dry algal mats and 10 glacial dust samples were collected from different dry lake sites and along the margin of polar ice cap, valleys and Nunataks, in a transect at an interval of 200-500m in and around Schirmacher Oasis.
3. Water samples from six different glacial lakes, three fresh snow samples and three blue ice samples were collected for palynological as well chemical analysis.

4. A lake bottom sediment profile (50cm.) was procured by using HYDROBIOS (Kiel Germany) gravity corer, from Long Lake (70 deg.45'20"S and 11 deg. 40' E) for palynostratigraphical studies.

Aerosol Radiative Forcing Measurement by Physical Research Laboratory, Ahmedabad.

Objective

1. To determine Characteristics of aerosols in Antarctica and the influence of Anthropogenic inputs on it.
2. Estimate the direct aerosol radiative forcing in terms of the reduction in the surface reaching solar flux
3. To compare the regional scale differences in direct aerosol radiative forcing in order to understand its relevance to the global climate change pattern.

Outcome

1. Ground based Ultraviolet (GUV) Radiometer was installed on a mound near Arravali Hut on a stand. Care was taken to place sensors at an higher elevation then the surrounding structures.
2. Quartz Crystal Microbalance (QCM) sampler too was installed at the Arravali hut with the inlet tube placed outside the hut.
3. Data from the GUV Radiometer were automatically downloaded on to the computer. Daily backup of this data was taken on floppies.
4. QCM, a manually operated instrument was operated on all fair weather days.
5. Besides this the two other instruments used to obtain data at regular intervals were Sunphotometer and Microtops II

Results

Results obtained from all the instruments are to be analysed at the Physical Research laboratory, Ahmedabad.

D. Biological Sciences and Environmental Conservation

Ecology of Microplankton by Dr I. K. Pai of Department of Zoology, Goa University, Goa.

Outcome

1. In order to study the ecology of lakes of Schirmacher oasis, 16 lakes of various sizes were subjected to the studies. Of which, seven are located to NW, 5 to west, one to NE, one to SE and one to South of Maitri station.
2. Depending on the size of the lake, one or two sites in each lake were selected and at each site two samplings were undertaken.
3. As preliminary work, various Physico-chemical parameters were analysed by using water testing kits of C.P.R. Foundation, Chennai. Further analyses of these water samples will be carried out at a later stage, at Goa University.
4. To analyse the fauna of these lakes, 25 lit. of water at each sampling site was filtered through an appropriate mesh and the concentrated to 300-500ml. quantity and they were narcotised and preserved in 4% formaldehyde solution. Actual identification, their density, distribution etc., will be undertaken at Department of Zoology, Goa University, Goa at a later stage.
5. Other objectives, as mentioned in the proposal will also be fulfilled at the institute.

E. Human Biology and Medicine

Leptin neuropeptide status by Praveen Vats of Defence Institute of Physiology and Allied Sciences (DIPAS), Delhi

Objective

The study was undertaken to assess the changes in leptin and neuropeptide Y levels in human volunteers at Antarctica and their role in regulation of appetite, food intake and change in body weight.

Outcome

1. The study was conducted in three phases viz., Phase I at Goa, Phase II immediately after reaching Antarctica and Phase III after one month of stay at Antarctica. Blood, urine and food samples were collected, apart from anthropometric and body fat measurements. Further, test for taste sensitivity and perception was carried out during the expedition period.

2. Blood samples will be analysed at DIPAS, for plasma leptin, NPY, insulin, triglyceride, cholesterol (total, HDL and LDL) protein and for thyroid function test.
3. Food samples will be analysed for various nutrients and energy intake.

F. Engineering and Communication

Work done by Anil Sonawane of Research and Development (Engineering) Establishment, Pune.

Objectives

Feasibility studies for relaying the hot water pipeline of the central heating system at Maitri.

Outcome

Attempted to prepare P & I diagrams of the existing pipeline. This exercise was aimed at making the process of changing the pipeline trouble free. No attempt could be made to change the pipeline as the pipe lines were to be shifted to Maitri by convoy.

Maintenance of communication linkages from Maitri and up-gradation of communication equipment at Maitri by N S Shekhawat, Bhagawati Prasad and K N Khanduri of Defence Electronic & Application Laboratory (DEAL), Dehradun

Outcome

1. INMARSAT 'B' (Skanti): Inmarsat 'B' terminal supplied by Skanti was installed at the Girnar hut. All four services linked to the terminal were checked and were found to be working satisfactorily. The e mail facility through VSNL could not be established.
2. INMARSAT 'B' (Sailor): Inmarsat 'B' terminal supplied by Sailor was commissioned at the Radio Room of Maitri station. All facilities attached to the terminals were checked and found working properly.
3. INMARSAT 'B' (Skanti): Inmarsat 'B' terminal of Skanti, sent to Matri along with the 19th expedition was tested at Girnar hut. This terminal appeared to be functioning normally but was not put in use as VSNL had cancelled its registration due to non use.

4. INMARSAT 'C (Sailor): Inmarsat 'C terminal was commissioned at the Radio Room of Maitri station. Telex transmit/receive function of terminal as well as the e mail transmission appeared to be functioning normally. However in the existing set up at Maitri the C terminal was of limited use.
5. Weather Fax Machine: Two weather fax (FX-220) machines were installed and checked and were found working. One machine was installed at the Radio room of Maitri station.
6. Periodic Antenna (LPA): LPA (14 MHz to 30 MHz) could not be erected and commissioned.

Wind Energy Utilization in Antarctica by Kanakamutthu and J Narayana of National Aerospace Laboratory (NAL), Bangalore.

Objective

1. Installation and commissioning of Vertical axis Windmill, charger, signal conditioner and data logger.
2. Conduct experiment to collect machine output data and wind data.
3. Installation and commissioning of portable windmill in Sankalp.
4. Conduct power consumption study.

Outcome

1. Successfully completed the installation and commissioning of the portable Windmill on a field living module 'Sankalp' for charging the Battery.
2. Successfully completed installation and commissioning of prototype vertical axis Windmill near the Nandadevi hut. This was done on experimental basis to check its performance under Antarctic condition.
3. Carried out Maintenance of the windmill battery charger installed near the communication repeater station at Veteiah nunatak.
4. Installed a small battery charging windmill near Maitri workshop.
5. Collected Power consumption data at the Indian Antarctic Base.

Scientific Work Carried out During Wintering

India Meteorological Department: Shri Suresh Kumar and M. K. De from MD wintered over during the period in Antarctica. They had a task

to study various Meteorological parameters and Ozone hole phenomenon including surface and upper air observations (for radiation and vertical profile of ozone). The entire assigned scientific task was completed successfully. The Ozone was monitored through Brewer Spectrometer on all clear weather days. Highlighted below are the minimum and maximum Meteorological parameters.

1. **Pressure:** The highest pressure recorded was 1011.4 mb at 00 Z on 25th May, 2001 and the lowest pressure recorded was 944.2 mb at 06 Z on 19th September 2001. April 2001 was the month with lowest mean average pressure at 979.5 mb while June 2001 was the month with highest mean average pressure at 990.3 mb.
2. **Temperature:** The highest temperature recorded was +7.4 degrees centigrade at 12 Z on 24th December 2001 while the lowest temperature recorded was -31.8 at 18 Z on 28th July 2001. The mean annual temperature was 9.8 degrees centigrade. January was the warmest month with temperatures averaging to +0.3 degrees while August was the coldest with average temperature of -19.2 degrees centigrade.
3. **Wind:** Maximum wind speed of 65 knots was recorded at 09 Z on 26th June 2001 and the maximum gust of 84 knots was recorded at the same time.
4. **Snow fall & Sky condition:** A total of 19.3 mm of snowfall was recorded during the 33 days with snowfall. September was the month with maximum snowfall. 17 blizzards were experienced during the year covering a span of 34 days of the year. October emerged a month with maximum blizzards. December 2001 was bright and sunny with minimum amount of cloudiness.
5. Maximum number of Auroras was seen in October 2001.
6. Concentration of Ozone peaked at 321 DU on 11th January 2002 from its minimum value of 113 DU recorded on 29th September 2001. The depletion was of the order of 60% from its maximum.

Indian Institute of Geomagnetism: Shri Sumendra Singh wintered over from IIG. The task he had on hand was monitoring the magnetosphere - ionosphere interactions over Schirmacher Oasis, Antarctica using fluxgate magnetometer and Rio meter. These observations were carried out without any interruptions. Besides this, atmospheric electrical environments related to various meteorological weather conditions was studied using AD 549 electrometer with long wire antenna. This new experiment on

atmospheric current was carried out successfully and it fulfilled the objective of finding out suitability of Maitri station for such experiment.

National Geophysical Research Institute: Shri L Prem Kishore from NGRI wintered over. The two tasks he had were in the fields of tele-Seismic Studies and Planetary Geodetic Studies.

Tele-Seismic Studies: The operational Seismic Station at Maitri monitors the Seismic activity within Antarctica and also aims to monitor tele-seismic activity as well as the activities in the mid oceanic region of Indian Ocean. It exchanges vital data on seismic activity and also envisages participation in global earthquake epicenter determination program. The Seismic station is equipped with high resolution digital seismograph station as well as analog seismic recorder. During the year the station was up graded with installation of additional Broad Band system with Gurlap CMG-3ESP sensor and digital acquisition system RT 121 from Reftek. The Seismic Vault was opened for installation of Broad Band Sensor and was found intact without any ice within the vault. The Experience of earlier wintering indicated that there was a considerable drop in the temperature within the hut and so a wooden partitioning was made to isolate the bay containing equipment within the Tirumala hut. The observations were taken without any interruption. Continuous monitoring of Seismic data was done and data was stored in CD's for further analysis. During the year about 350 earth quakes have been recorded at Maitri. The significant events recorded during the year were:

26th Jan 2001 arrival time at Maitri of the seismic wave was at 03 Hrs 30 minutes and 30.83 Seconds. This was a devastating earthquake that affected parts of Kutch and Ahmedabad in Gujrat.

24th Oct 2001. Arrival time at Maitri of the seismic wave was at 22 Hrs 42 Minutes and 45.45 Seconds. This is the nearest event recorded, which was around 50 Kms from Maitri in the north Eastern direction.

Planetary Geodetic Studies: The purpose of this permanent station is to measure very long base line between India and Antarctica by continuous GPS monitoring, to provide accurate position in the global reference frame, as also a fundamental marking for relative positioning. It also enables to study the tectonic movements i.e. movement of Antarctic plate with reference of other plates. Having established a permanent GPS station, NGRI

could participate in the Global Antarctic Program Campaign by contributing GPS data as a fundamental marker to the SCAR campaign 2001. During past one year continuous GPS monitoring and data collection was carried out with the help of Ashtech Z XII GPS Geodetic receiver till 8th February 2002. The ASHTECH receiver was set up everyday and allowed to acquire data throughout 24 hours with set interval of 15 seconds. Since 9th February 2002 the old receiver has been replaced with TURBOROGUR GPS Receiver System. The raw data collected has been converted to standard RINEX format and stored on CD's for further processing.

Geological Survey of India: Mervin D'Souza & Amit Dharwadkar from GSI wintered over. The task assigned was to continue glaciological and permafrost observations for the entire year. The team also had a plan of carrying out thematic mapping in parts of Wohlthat Ranges during winter. As four vehicles were down for repairs, so going to mountains using vehicles did not materialize. The opportunity was however utilized to do extensive theme specific sampling of orthogneisses and meta ultramafic rocks from Schirmacher oasis. About 75 samples were collected for detailed geochemical studies of orthogneisses and metabasics. The distant nunataks SW of Schirmacher Oasis was geologically mapped in detail and 30 rock samples from these nunataks were collected for petrochemical and geochemical studies. Daily permafrost observations were taken and the stake network at DG was monitored at regular interval.

An Appraisal and Recommendations

The Expedition and the wintering was a great experience. We had our share of successes and also disappointments. The harsh realities of Antarctica made us wiser. We have gained some experience and have made some contribution to make Maitri a better place to live. In the lines that follow I have attempted to highlight the task we completed and things we think should be done to make the Indian Expedition to Antarctica better than what it is.

1. The entire ship was vacated within 6 days of reaching Antarctica. The Helicopters till the 19th expedition were operated from the Ship as the base, for the first time the Helicopters were located during 20th IAE and operated from Maitri. This resulted in better planning of field related activity.

2. Installation of T' type antennae for VLF experiment during summer for the Barkatullah University, Bhopal was immediately done on reaching Maitri. The copper wire brought for the antenna was the regular copper wire available in the market with plastic jacket on it. It had to be peeled prior to fixing on to the antenna and apparently did not have the required strength to withstand strong Antarctic winds. The university should have sent the special copper wire strong enough to withstand Antarctic winds
3. Commissioning of 2 New SATCOM units was completed within 15 days of reaching Antarctica. The units are functioning satisfactorily.
4. Maximum amount of helicopter time utilized for scientific field work. Full logistic support had been extended to the German scientists and the GPR experiment. The task has been completed to the total satisfaction of the German scientists associated with the project. They could stick to the date wise schedule they had given for completion of the work. Logistic support was also extended for establishing camp at Gruber Mountains. About 40 hrs of helicopter time was utilised exclusively for this work
5. Survey of Ice Shelf coastline done by NHO and GSI to find an alternate site for disembarking the cargo. The suggestions on alternate site for berthing submitted to NCAOR on return
6. Hot water pipeline replaced in the Maitri station without compromising on station heating during winter. The task of changing pipeline completed within 24 hours
7. Internal lobby, Lounge and Kitchen area repainted. The stilts painted with red oxide. The exterior of Maitri station painted.
8. Removed three container load of junk lying around Maitri station for backloading.
9. Cleaning of Maitri and its surrounding was followed as a routine during wintering.
10. Installation of Klargesters as a task was not taken up as it was found **not** to be practical. The two Klargesters are already installed and setting up a third Klargesters could cause cluttering and unnecessary snow accumulation in the area.

Recommendations

1. The night soil generated at Maitri is burnt everyday and collected in barrels. Alternate systems should be installed in consultation with the other Antarctic Stations.

2. Managing the wastewater generated in the kitchen, bathrooms and urinal is a major concern. A full proof Waste water treatment plant should be installed within a heated interior at the existing location.
3. Painting the exterior walls of Maitri should be done regularly.
4. New snow vehicles should be inducted at Maitri.
5. At least one high mobility crane should be sent to Maitri which can function as recovery vehicle.
6. New summer living modules with all facilities should be created at the Indian Base Maitri.
7. The Scientific component of the expedition should reach Antarctica by November first week and return by February first week. This can be achieved as an air route to Maitri is functional. The Wintering team can be sent to Maitri in March along with the cargo. Field work can be done using vehicles. Helicopter support too can be obtained if necessary. Experience has shown that November December and January are the best months to work in Antarctica.

During 18 WOT the wintering members were asked to give suggestion based on their experience to make Maitri a better place to live. Many practical suggestions were made by the members some of which are reproduced here.

Suggestions made by the Members of 18th WOT

1. Need to have very good Ice Shovels at Maitri. The Ice shovels supplied were not suitable for Shoveling Antarctic Ice. With frequent blizzards in Antarctica a need for a very good shovel is very important. Sourcing good quality ice shovels from Countries living with perpetual ice conditions could be explored.
2. Spanners and other tools. Working with metal workshop tools during winter outside exposes a person to cold burns. Some of these metal tools also break due to the metallurgy of the metal not being suitable to withstand Antarctic cold. It could be helpful if the tool handles have a PVC/Rubber coat to minimize the risk of cold burns.
3. Induction of two more PB330 or better version vehicles to augment the present haulage capacity.
4. Proper training of personals on existing models of vehicles in India. The training presently is restricted to rudimentary exposure to a

vintage model PB 170 available at EME School, Vadodara. The present models available at Maitri i.e PB 330 & PB 270 are totally different and hence the training on PB 170 becomes redundant.

5. Spares parts catalogues of all the equipment being imported into Maitri should be available at Maitri. Ordering for correct spare parts will be easy then.
6. A living module is always carried with the convoys. The present living module 'Banjara' is not suitable for living, as it cannot be heated up or is not exactly suited to meet other daily needs if the convoy is stuck in severe blizzard. Being stranded in Banjara for few days apparently turns into a nightmare. Suggestions were made to induct a self-contained living module for convoy on the lines of one held by the Russians.
7. Working overalls and similar good quality inners are must for wintering. The existing wintering team preferred to use old and badly worn out HH overalls as they were very practical to meet the Antarctic needs and also comfortable to wear and work.
8. Exhaustive training on installed Thermax Boiler system and water cooled 125 KVA Crompton generators at the company. Apparently the persons inducted to maintain this life support system have never seen such systems earlier.
9. The electric cables used here-specially the ones exposed to Antarctic climate do not sustain the Antarctic conditions for longer period. The power cables are laid out in open and there is always a risk involved as they open up. We could explore a possibility of inducting good quality power cable, which could last longer in Antarctic conditions.
10. The generators are housed in containers, which tend to vibrate when the generators are working. Vibrations effect the health of the generators and all effort be made to minimize this prior to installation.
11. There should be a good heated bay created in the vehicle workshop. This will enable the persons to carry out major repairs even during winter. A hanger should be made at Maitri for vehicles, as during winter all the vehicles are buried in ice. Removing ice after that could cause damage at times to the sensitive parts.
12. There is a Libher Crane available at Maitri. The engine is old and needs to be overhauled. If this is done it can be utilised effectively for routine work at Maitri.

13. There is lot of equipment like pumps etc lying at Maitri. The equipment can be sent for repairs and then resent to Maitri.
14. Possibility of overhauling the PB270 & PB330 vehicles on main land be explored.
15. There were frequent requests for increasing the telephone time allotted to each member. Suggestions included that a fixed amount in dollars be allotted to each member, which can be utilised to its optimum value to the member's advantage.
16. The doctors should get training in handling the pathological laboratory equipment available at Maitri. Also possibilities for augmenting the existing medical facilities can be explored. Medicines being sent to Maitri should be freshly packed, as it will optimise its use in terms of expiry date.
17. IMD is in need for balloon launching facility.
18. Maitri mail handling facility at NCAOR may be created. The mails received from families should be promptly sent to Maitri. Being in touch with family back home always keeps the team in good spirits even under extreme Antarctic conditions.
19. The quality of marine ply being sent to Maitri can be improved.
20. Refurnishing of Maitri Lounge can be undertaken. The need is for realistic strong heavy duty furniture.

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Fig.1: The Expedition Vessel M/v Magdalena Oldendorff had a smooth sailing to Antarctica.



Fig. 2: Ship in the beautiful Antarctic waters.



Fig.3: Ship experiencing snow storm.



Fig.4: Helicopters parked at Maitri. It is the first time that the helicopters were parked at Maitri for 16 days and were operated from Maitri for Scientific work.



Fig.5: A field camp established in Gruber Mountain by the University of Dresden, Germany team.



*Fig.6: IIG & GSI field camp at **Nunatak** Svodene 150 miles away from Maitri.*



Fig.7: University of Dresden geodetic survey point in Humboldt Mountains, Wohlthat Range, CDML.



Fig.8: Geological field work in progress in the Muhlig-Hoffmannfjella by GSI.



Fig. 9: Unusual fast ice conditions in India bay region. The ship had to wait for the fast ice to crack till 1st March 2001 to berth.



Fig. 10: Ship comes along side to the shelf to offload cargo.



Fig. 11: Ship on shelf-offloading continued even during the night.



Fig. 12: Cargo on shelf being readied to be taken to Maitri by land convoys.



F/g. 75; Maitri and its surrounding turns into a snow dumping ground during winter.



Fig. 14: Generator area covered with ice during winter.



Fig. 15: A view of summer hut area during winter.



Fig. 16: Maitri and surrounding appears totally different during summer.



Fig. 17: 18" WOT Painting Maitri station.



Fig. 18: Vehicles buried in ice during polar nights being cleared.



Fig. 19: Norwegian Expedition land at the Russian Air strip in December.



Fig. 20: 18th WOT members waiting for the 21st Indian Antarctic Expedition at the helipad.



Fig. 21: Indian Antarctic Base Maitri standing tall



Fig. 22: Adios Antarctica!